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SI—06—2025

FACULTY OF SCIENCE AND TECHNOLOGY

BCA (Second Year) (Fourth Semester) EXAMINATION

APRIL/MAY, 2025

(CBCS/Revised Pattern)

COMPUTER GRAPHICS

(Thursday, 3-4-2025) **Time : 2.00 p.m. to 5.00 p.m.**

Time—3 Hours *Maximum Marks—75*

N.B. :— (*i*) *All* questions are compulsory.

(*ii*) Figures to the right indicate full marks.

(*iii*) Assume ;suitable data, if required.

(*iv*) Use of any electronic media such as mobile phone, digital diary and electronic calculator is not permitted.

1. Attempt any *five* of the following (3 marks each) : 15

(*a*) What is Graphical user interface ? Explain.

(*b*) What is rotation ? Explain.

(*c*) Explain symbols and instances in brief.

(*d*) Discuss working of Cathode Ray Tubes.

(*e*) What is Scaling ? Explain.

(*f*) Define the terms : Line and Line Segement.

(*g*) What is Plotter ? Explain.

P.T.O.

2. Attempt any *three* of the following (5 marks each) : 15

(*a*) What is Computer Graphics ? Discuss application areas of Computer Graphics in brief.

(*b*) Explain Bresenham's line algorithm in brief.

(*c*) Explain posting and unposting segments in brief.

(*d*) Discuss ground rules for graphics s/w design. Explain.

(*e*) Differentiate between Random scan displays and Raster scan displays.

3. Attempt any *three* of the following (5 marks each) : 15

(*a*) What is two-dimensional transformation ? Explain translation with example.

(*b*) Explain in brief segment naming scheme.

(*c*) Explain Polygon clipping algorithm.

(*d*) Discuss Windowing function in brief.

(*e*) Discuss working of Light Pen in brief.

4. Attempt any *three* of the following (5 marks each) : 15

(*a*) Explain direct view storage tube in brief.

(*b*) Explain digitial differential algorithm in brief.

(*c*) Explain Reflection with example.

(*d*) Explain function domains in brief.

(*e*) What is segment table ? Explain.

5. Write short notes on any *three* of the following (5 marks each) : 15

(*a*) Joystick

(*b*) Translation

(*c*) Shear

(*d*) End point codes

(*e*) Default error conditions.

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